

PSM

Question-Engraved Concept Book
NEET PG · INICET · FMGE · UPSC CMS

■ Based on 260+ PYQs • 2015–2025 • 10 High-Yield Chapters

Concepts emerge from questions. Patterns replace memorization.

Preventive & Social Medicine (PSM / Community Medicine)

How To Use This Book

This is NOT a textbook. This is a **Question-Engraved Concept Book**. Every concept grows *out of* questions — not beside them. As you read, you will repeatedly feel: **'Now I understand WHY the examiner asked this.'**

Usage Mode	Instructions
■ First read	Read each chapter as a narrative. Don't skip the traps section.
■ Second pass	Cover answers, attempt questions, then compare your reasoning.
■ Rapid revision	Use only the Revision Capsule at each chapter end.
■ Exam day	Re-read only the Trap Alerts and Memory Hooks.

■ The Secret of High Scorers

High scorers don't memorise more — they understand patterns. Every PSM PYQ in the last 8 years fits ≤ 15 core conceptual clusters. This book maps all of them. Once you see the patterns, you stop fearing PSM.

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Biostatistics

The Examiner's Playground — 30+ Questions in 8 Years

■ What the Examiner REALLY Targets

Biostatistics is the single most tested PSM sub-topic. The examiner doesn't test formula derivations — they test your ability to **choose the right tool** for a given data scenario. The four most-repeated decision points are: (1) which test to use, (2) what SD/distribution tells you, (3) how to read screening test calculations, and (4) how to define bias.

Core Concept 1: Choosing the Right Statistical Test

Think of statistical tests as *tools for different jobs*. The examiner gives you a scenario and expects you to identify which tool fits. The decision tree is simple once you internalize one rule: **First ask: what kind of data is this?**

Scenario	Correct Test
Two independent group means (continuous data)	Unpaired T-test
Same group before & after (continuous)	Paired T-test
Comparing proportions / categorical data	Chi-square (χ^2) test
≥ 3 group means (continuous)	ANOVA (F-test)
Predict one variable from another	Regression coefficient
Measure strength of association	Correlation coefficient (r)
Compare variability between datasets	Coefficient of Variation (CV)
Non-parametric data (ordinal/skewed)	Mann-Whitney U / Kruskal-Wallis

Q: A study assesses malnutrition in rural (30/100 malnourished) and urban (20/100). Which test?

- A. Paired t-test
- B. Chi-square
- C. Unpaired t-test
- D. ANOVA

✓ **B. Chi-square**

The data is PROPORTIONS (malnourished vs not) in two INDEPENDENT groups. Proportions/categorical data → Chi-square. Trap: students pick t-test because two groups are mentioned — but t-test is for MEANS (continuous data), not proportions. Paired t-test = same group before/after. This is rural vs urban = independent groups.

Q: To compare haemoglobin levels between two independent groups, which test is used?

- A. Chi-square
- B. Unpaired t-test
- C. Paired t-test
- D. ANOVA

✓ **B. Unpaired t-test**

Haemoglobin is a CONTINUOUS variable. Two INDEPENDENT groups = Unpaired t-test. Paired t-test = would be used if the SAME patients had Hb measured before and after treatment. Chi-square = categorical data only (e.g., anaemic vs not).

■ The t-test vs Chi-square Trap

The #1 biostatistics trap: When you see 'two groups,' students instinctively jump to t-test. STOP. Ask: 'What is being compared — MEANS or PROPORTIONS?' Proportions, percentages, counts, categorical data → Chi-square ALWAYS.

Core Concept 2: Normal Distribution & Standard Deviation

The normal (bell) curve is symmetric. In a normal distribution, mean = median = mode. The **68-95-99.7 rule** is tested almost every year in some form.

Fact	Value / Interpretation
Mean $\pm$ 1 SD	68% of values
Mean $\pm$ 2 SD	95% of values (95.4%)
Mean $\pm$ 3 SD	99.7% of values
Beyond 2 SD	Statistically significant ($p < 0.05$)
Skewed right	Mean > Median > Mode
Skewed left	Mean < Median < Mode
Best central tendency in skewed data	Median (not affected by outliers)

Q: A study has mean = 200, SD = 20. 68% of values lie between:

- A. 160–240
- B. 170–230
- C. 180–220
- D. 190–210

✓ **C. 180–220**

Mean $\pm$ 1 SD = $200 \pm 20 = 180\text{--}220$ (68%). Mean $\pm$ 2 SD = $200 \pm 40 = 160\text{--}240$ (95%). Mean $\pm$ 0.5 SD = $200 \pm 10 = 190\text{--}210$ (38%). Trap: Option A (160–240) looks like '2 SD range' and many select it confused by the question asking for 68%.

Core Concept 3: Screening Tests — PPV, NPV, Sensitivity, Specificity

This is the most calculation-intensive area. The key is to **always draw the 2×2 table** before solving. Never try to solve from memory alone.

	Disease +	Disease –	Total
Test +	TP (a)	FP (b)	a+b
Test –	FN (c)	TN (d)	c+d
Total	a+c	b+d	

Measure	Definition / Key Fact
Sensitivity	TP/(TP+FN) — detects true positives
Specificity	TN/(TN+FP) — rules out true negatives
PPV	TP/(TP+FP) — if test +ve, chance of disease
NPV	TN/(TN+FN) — if test –ve, chance of no disease
Sensitivity ↑ with	Low cut-off / wide net — more FP, fewer FN
Specificity ↑ with	High cut-off / strict criteria — more FN, fewer FP
PPV depends on	Prevalence (PPV ↑ when prevalence ↑)
NPV depends on	Prevalence (NPV ↑ when prevalence ↓)

Q: CA-125 screen for ovarian CA: 60/100 test-positive had disease; 20/100 test-negative had disease. NPV?

- A. 20/100
- B. 40/100
- C. 60/100
- D. 80/100

✓ **D. 80/100**

Build the 2×2 table: TP=60, FP=40, FN=20, TN=80. NPV = $TN/(TN+FN) = 80/(80+20) = 80\%$. Trap: Many select 60/100 (the PPV) or 20/100 (the FN rate). Always remember NPV is about those who tested **NEGATIVE** — 80 out of 100 negatives truly don't have disease.

Core Concept 4: Types of Bias

Bias = systematic error. The examiner asks you to identify bias from a scenario. The trick is each bias has **ONE** unique feature that no other bias has.

Bias Type	Defining Feature
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Berkson's (Admission Rate) Bias	Hospital-based study; people with TWO diseases over-represented
Lead Time Bias	Screening finds disease early → survival APPEARS longer, but mortality unchanged
Length Bias	Screening catches slow-growing (less lethal) tumours more than fast-growing ones
Neyman (Incidence-Prevalence) Bias	Fatal/short-duration cases missed in prevalence studies
Recall Bias	Cases remember exposure better than controls (case-control studies)
Volunteer Bias	Volunteers are healthier/more motivated than non-volunteers
Observer Bias	Researcher's expectation influences measurement
Confounding Bias	Third variable associated with BOTH exposure and outcome — distorts association

Q: In a hospital study, patients with both breast cancer AND meningioma are more likely to be admitted than those with either alone. This inflates the apparent association. Which bias?

- A. Lead time bias
- B. Berkson's bias
- C. Neyman bias
- D. Recall bias

✓ **B. Berkson's (Admission Rate) bias**

Berkson's bias = HOSPITAL-BASED studies where people with 2 diseases (exposure + outcome) are MORE likely to be hospitalised → inflated association. Key: If the question says 'hospital-based case-control' + 'dual pathology over-admission' → Berkson's.

Q: Screening detects cancer at Stage I instead of Stage III. Five-year survival improves, but mortality rate doesn't change. This is an example of:

- A. Length bias
- B. Selection bias
- C. Lead time bias
- D. Observer bias

✓ **C. Lead time bias**

Lead time = the interval between early detection and the time diagnosis would have been made clinically. Survival appears improved because the clock starts earlier — not because the disease course changed. The confirming clue is: 'mortality didn't change.' If mortality changed, it would be a true benefit.

■ Confounding vs Effect Modification

Confounding: A third variable distorts the exposure-disease relationship. It is ASSOCIATED with both exposure and outcome AND unequally distributed. Trap: If the question says 'equally distributed' → NOT a confounder (it cancels out). Key definition: 'Associated with BOTH exposure AND disease, distributed UNEQUALLY.'

Core Concept 5: p-value, Null Hypothesis, and Statistical Significance

The p-value is the probability of getting your result IF the null hypothesis is true. $p < 0.05$ = statistically significant = reject null hypothesis. This does NOT mean clinical significance.

■ Reading p-values in MCQs

When a question says 'p = 0.03' → significant, reject H_0 , the groups ARE different. When 'p = 0.08' → NOT significant, cannot reject H_0 . Examiner trap: 'p < 0.05 means the result is clinically important' — FALSE. Statistical significance ≠ clinical significance.

■ ULTRA HIGH-YIELD REVISION CAPSULE

- Chi-square = categorical/proportions | Unpaired T = two independent means | Paired T = before-after means
- 68%-95%-99.7% rule: $\pm 1SD$, $\pm 2SD$, $\pm 3SD$
- Skewed data → Median is best central tendency
- NPV = $TN/(TN+FN)$ | PPV = $TP/(TP+FP)$ | Sensitivity = $TP/(TP+FN)$ | Specificity = $TN/(TN+FP)$
- PPV ↑ with prevalence ↑ | NPV ↑ with prevalence ↓ (sensitivity/specificity don't change with prevalence)
- Berkson's = hospital + 2 diseases | Lead time = early detection + no mortality change
- Confounding = associated with BOTH + distributed UNEQUALLY
- $p < 0.05$ = reject null hypothesis = statistically significant
- Coefficient of Variation (CV) = compare variability between datasets with different units
- Coefficient of regression = predict one variable FROM another

Epidemiology & Study Designs

Which Study for Which Purpose — A Decision Framework

■ What the Examiner Targets

The examiner gives you a research SCENARIO and asks you to identify the study design. The key to every question is: **What is the unit of analysis — individual or population? And in which direction does time flow?**

Time Direction / Design Feature	Study Type
Exposure → Disease (present to future)	Cohort study → Relative Risk (RR)
Disease → Look back at exposure	Case-control → Odds Ratio (OR)
Both exposure & disease measured NOW	Cross-sectional → Prevalence
Aggregate data (regions/time periods)	Ecological → Risk of ecological fallacy
Intervention assigned by investigator	Experimental (RCT)
Natural experiment / assigned by nature	Quasi-experimental
Individual followed over time from past	Retrospective cohort

Q: A researcher collects lung cancer death data from hospitals and cigarette sales from the taxation dept in the same period. What study type?

- A. Cross-sectional
- B. Ecological
- C. Cohort
- D. Case-control

✓ **B. Ecological study**

KEY: The data is aggregate/group-level (population-level cancer rates vs population-level cigarette sales). No individual-level data is collected. Unit of analysis = population (not individuals) → Ecological. Warning: ecological studies are subject to ECOLOGICAL FALLACY — a relationship seen at population level may not hold at the individual level.

Q: Doctors in 1970 were asked about their smoking habits. They are followed until 2000 to record lung cancer occurrence. Design?

- A. Case-control
- B. Cross-sectional

- C. Prospective cohort
- D. Retrospective cohort

✓ **C. Prospective cohort**

Exposure (smoking) identified FIRST, then followed forward in time for outcome (lung cancer). This is the classic PROSPECTIVE COHORT design (Doll & Hill study). Generates: Relative Risk (RR), Attributable Risk (AR). Retrospective cohort = exposure data collected from PAST records, then outcome tracked.

Measures of Association: RR, OR, AR

Measure	Study + Formula / Use
Relative Risk (RR)	Cohort study. Risk in exposed / Risk in unexposed
Odds Ratio (OR)	Case-control. Cross-product ratio (ad/bc). $\approx$ RR when disease is rare
Attributable Risk (AR)	Incidence exposed – Incidence unexposed. Absolute excess risk
Population AR (PAR)	Incidence total – Incidence unexposed. Public health importance
NNT	1/ARR. Number needed to treat to prevent one outcome

Q: Cross-product ratio (Odds Ratio) is obtained from:

- A. Ecological study
- B. Cohort study
- C. Cross-sectional
- D. Case-control

✓ **D. Case-control study**

OR = $(a \times d) / (b \times c)$ from a 2x2 table. Used in CASE-CONTROL studies because you start with disease status (cases vs controls) and look backwards at exposure. Trap: Some say 'OR can be calculated from a cohort study too' — TRUE, but the question asks about the PRIMARY measure from case-control studies. OR $\approx$ RR only when disease prevalence is low (<10%).

Confounding — The Full Picture

A confounder is a variable that creates a false association between exposure and disease. It must satisfy THREE criteria: (1) Associated with the EXPOSURE, (2) Associated with the DISEASE (independent of exposure), and (3) NOT on the causal pathway between exposure and disease.

Q: A confounding factor is defined as one which is associated with:

- A. Exposure only, distributed unequally
- B. Both exposure AND disease, distributed unequally

C. Both exposure AND disease, distributed equally

D. Disease only, distributed unequally

✓ **B. Associated with BOTH exposure and disease, distributed UNEQUALLY**

ALL three words matter: (1) BOTH = it must link to exposure AND disease. (2) UNEQUALLY = if it were equally distributed between groups, it would cancel out and not confound. Trap: Option C has 'equally' distributed — that would NOT cause confounding. Option A says 'exposure only' — that's just a risk marker, not a confounder.

Surveillance: Types and Purpose

Type	Key Feature
Active surveillance	Health workers actively search for cases (field visits, phone calls)
Passive surveillance	Routine spontaneous reporting by providers to health authorities
Sentinel surveillance	Selected high-quality reporting sites monitor trends in specific populations
Syndromic surveillance	Monitor non-specific symptoms (ED visits, OTC sales) for early outbreak signal
Surveillance objective	Detect outbreaks/trends EARLY for timely public health response
Monitoring	Routine, continuous measurement of programme inputs/outputs (≠ surveillance)

■ ULTRA HIGH-YIELD REVISION CAPSULE

- Cohort → RR | Case-control → OR | Cross-sectional → Prevalence
- Ecological = population-level data, risk of ecological fallacy
- Prospective cohort = exposure identified FIRST, followed forward
- Retrospective cohort = exposure from PAST records, outcome from records
- $OR \approx RR$ when disease prevalence < 10% (rare disease assumption)
- Confounder: associated with BOTH + unequally distributed
- Attributable Risk = Incidence(exposed) – Incidence(unexposed)
- Active surveillance = health workers search actively
- Passive surveillance = routine reporting by providers
- Sentinel surveillance = selected sites monitoring specific populations

Screening & Levels of Prevention

Leavell & Clark's Framework — Tested Every Year, Multiple Angles

■ What the Examiner Targets

This is the second most-repeated PSM conceptual cluster. The examiner tests you in two ways: (1) Give a clinical intervention and ask which LEVEL of prevention, and (2) Give a screening scenario and ask about validity/test properties. Both require a firm conceptual map, not memorisation.

The 5 Levels of Prevention (Leavell & Clark)

Level	Definition & Examples
Primordial Prevention	Prevent risk factors from emerging in the population (e.g., tobacco-free generation, healthy school lunch policies)
Primary Prevention Level 1: Health Promotion	General health improvement not targeting a specific disease (e.g., sanitary latrines, adequate nutrition, health education)
Primary Prevention Level 2: Specific Protection	Protection against a specific disease (e.g., immunisation, OCP, seatbelt use, iodised salt, fluoride)
Secondary Prevention: Early Diagnosis & Treatment	Screening tests, clinical examination to detect pre-symptomatic disease. Reduce severity/duration.
Tertiary Prevention: Disability Limitation	Prevent worsening/disability in established disease (e.g., callipers in polio, preventing limb loss in diabetes)
Tertiary Prevention: Rehabilitation	Restore function after disability (physiotherapy, prosthetics, vocational rehab)

Q: Vitamin A prophylaxis in children is which level of prevention?

- A. Primordial
- B. Specific protection (Primary)
- C. Early diagnosis
- D. Disability limitation

✓ **B. Specific protection (Primary Prevention Level 2)**

Vitamin A prophylaxis prevents a SPECIFIC disease (Vitamin A deficiency / xerophthalmia). Any intervention that prevents a specific condition = Specific Protection. Contrast: Balanced diet (general nutrition) = Health Promotion. Vitamin A treatment for a deficient child = Disability Limitation.

Q: Providing callipers (leg braces) to a child with poliomyelitis. Level of prevention?

- A. Health promotion

- B. Specific protection
- C. Disability limitation
- D. Rehabilitation

✓ **C. Disability limitation**

The disease (polio) has already occurred and caused damage. Callipers PREVENT FURTHER DISABILITY from the existing disease. Disability limitation = preventing worsening of existing disease. Rehabilitation = RESTORING function. Callipers support function but don't restore it.

Q: Seatbelt use while driving. Level of prevention?

- A. Primordial
- B. Health promotion
- C. Specific protection
- D. Early diagnosis

✓ **C. Specific protection**

Seatbelt targets SPECIFIC protection against road traffic injury. Common trap: students say 'health promotion' because it seems general. Key rule: if it targets a SPECIFIC hazard → Specific Protection. Primordial = prevent the risk factor from emerging at population level (e.g., policy banning drunk driving).

■ **The OCP Trap**

Oral contraceptive pill (OCP) = SPECIFIC PROTECTION (Primary Prevention Level 2) — it specifically prevents pregnancy. NOT Primordial. NOT Health Promotion. This is a classic distractor because contraception feels like 'general health.'

Lead Time & Screening Biases

Lead time is the period between screen-detectable disease and clinically symptomatic disease. Screening advances the diagnosis by the lead time. If the patient's prognosis doesn't change, survival from diagnosis appears longer — but this is an ARTEFACT (Lead Time Bias).

Q: 5-year survival for lung cancer improves after low-dose CT screening is introduced, but annual mortality from lung cancer doesn't change. This is due to:

- A. Selection bias
- B. Length bias
- C. Lead time bias
- D. Recall bias

✓ C. Lead time bias

The definitive clue: MORTALITY DID NOT CHANGE. Survival improved only because the diagnosis clock started EARLIER. Length bias would mean the screened tumours are inherently less lethal (slow-growing). Length bias clue: 'screened population has less aggressive tumours.'

Wilson & Jungner Screening Criteria (Classic 10)

These criteria define when screening is worthwhile. The examiner asks which condition DOES NOT satisfy screening criteria.

Wilson's Criterion	Meaning
Important health problem	High prevalence or serious condition
Accepted treatment available	Treatment must exist and be effective
Facility for diagnosis/treatment	Infrastructure must be available
Latent or early symptomatic stage	Screen BEFORE symptoms develop
Suitable test available	Sensitive, specific, acceptable to population
Test acceptable to population	Not overly painful/invasive
Natural history understood	Must know how it progresses
Policy on whom to treat	Clear criteria for follow-up/treatment
Cost-effective	Economic analysis must be favourable
Continuing process	Not a one-time event

■ Testicular Tumour — NOT Screened

Testicular tumours do NOT have a recommended population-level screening programme. They present as painless swelling and are diagnosed clinically. Compare: Breast (mammography), Cervix (Pap/HPV), Prostate (PSA), Colorectal (FOBT). Exam trap: All four above have screening — testicular CA does not.

■ ULTRA HIGH-YIELD REVISION CAPSULE

- Primordial = prevent risk factors from emerging at population level
- Specific Protection (PP Level 2) = seatbelt, vaccine, OCP, iodised salt, Vitamin A prophylaxis
- Health Promotion (PP Level 1) = sanitary latrines, nutrition, health education
- Early Diagnosis & Treatment = screening tests (Pap smear, mammography)
- Disability Limitation = prevent worsening (callipers, dialysis, insulin in DM)
- Rehabilitation = restore function (prosthetics, physiotherapy, vocational rehab)
- Lead time bias = survival ↑ but mortality unchanged → clue: 'mortality not reduced'

- Length bias = screened cases less aggressive → clue: 'screen-detected cases do better anyway'
- Screening NOT recommended for testicular tumours
- PPV↑ when disease prevalence↑ | Sensitivity/specificity = fixed test properties

Communicable Disease Control

NTEP, NVBDCP, HIV, Malaria — High-Frequency Pattern Recognition

■ What the Examiner Targets

The examiner does NOT expect encyclopaedic knowledge of every disease. They test SPECIFIC programme numbers, SPECIFIC surveillance indicators, and CLASSIFICATION rules. The three highest-yield disease domains are: TB (NTEP), Malaria (NVBDCP), and HIV.

Malaria: Surveillance Indicators

Indicator	Definition / Key Fact
ABER (Annual Blood Examination Rate)	% of population with blood smear taken in a year. Target $\geq 10\%$. Measures OPERATIONAL EFFICIENCY of surveillance.
API (Annual Parasite Index)	Confirmed malaria cases per 1000 population per year. Measures DISEASE BURDEN.
SPR (Slide Positivity Rate)	% of examined slides that are positive. Measures TRANSMISSION INTENSITY.
SFR (Slide Falciparum Rate)	% of positive slides that are <i>P. falciparum</i> . Measures SEVERITY risk.
ABER $\neq$ epidemiological indicator	ABER is OPERATIONAL efficiency indicator, not epidemiological indicator.
IRS rounds (API > 2)	2 rounds per year in high-transmission areas
Microscope: 1 DMC covers	25,000 population (NVBDCP norm)
OptiMAL test	Detects <i>P. falciparum</i> + <i>P. vivax</i> (pLDH-based RDT)

Q: Which malaria indicator measures the OPERATIONAL EFFICIENCY of the surveillance system?

- A. Annual Parasite Index
- B. Annual Blood Examination Rate
- C. Slide Positivity Rate
- D. Slide Falciparum Rate

✓ **B. ABER (Annual Blood Examination Rate)**

ABER = what PROPORTION of population had their blood examined. It tells you HOW HARD the programme is working (operational efficiency). Trap: API sounds like an 'efficiency' metric but it actually measures DISEASE BURDEN (cases per 1000). Examiner frequently asks: 'ABER is NOT an epidemiological indicator' — TRUE, it's an operational indicator.

NTEP (National Tuberculosis Elimination Programme)

Fact	Detail
MDR-TB definition	Resistant to at least RIFAMPICIN + ISONIAZID (2 most potent 1st-line drugs)
Pre-XDR-TB	MDR-TB + resistant to ANY fluoroquinolone
XDR-TB	MDR-TB + fluoroquinolone + at least one Group A drug (bedaquiline/linezolid)
GeneXpert (Xpert MTB/RIF)	Detects MTB AND rifampicin resistance within 2 hours
NIKSHAY	TB notification + patient support platform
DOTS incentive (DR-TB)	■500/month for DR-TB patients on treatment
Community TB detection	■500 incentive for community referral leading to DR-TB detection
Most peripheral RNTCP unit	Designated Microscopy Centre (DMC), not PHC or sub-centre
ESI extended sickness TB	Up to 2 years (ESI Act)

Q: MDR-TB is defined as resistance to:

- A. Only rifampicin
- B. Rifampicin + Isoniazid
- C. All 4 first-line drugs
- D. Rifampicin + any injectable

✓ **B. Rifampicin + Isoniazid**

MDR = Multi-Drug Resistant = resistant to the TWO most important first-line drugs. If only rifampicin resistant = RR-TB (treated like MDR-TB). XDR adds fluoroquinolone resistance. Pre-XDR = MDR + fluoroquinolone. Trap: 'All 4 first-line drugs' is not a definition; XDR has a specific additional drug class.

HIV: Surveillance, Diagnosis, Treatment

Topic	Key Fact
Sentinel surveillance purpose	Monitor HIV prevalence trends in specific risk groups (NOT identify individuals)
Diagnosis (asymptomatic adult)	2 ELISA tests (different antigens). NOT require Western blot
Opt-out HIV testing (pregnancy)	Standard of care — tested unless patient specifically declines
PEP initiation	As soon as possible, ideally within 2 hours, definitely < 72 hours
PEP regimen	TDF + 3TC + DTG for 28 days
PMTCT: highest risk factor for vertical transmission	High maternal viral RNA load
Elective C-section benefit	Reduces vertical transmission by ~50% (before labour + membrane rupture)

HIV + active TB	Start ATT FIRST, then ART within 2 weeks (all CD4 counts)
Exception (HIV + TB meningitis)	Delay ART 4–8 weeks to reduce paradoxical IRIS
IgA not transplacental	Secretory IgA (OPV antibody) does NOT cross placenta

Q: HIV + active TB patient (CD4=250). What to do?

- A. Start ATT first, then ART within 2 weeks
- B. Start ART first, then ATT after 2 weeks
- C. Start both simultaneously
- D. Start ATT first, then ART after 8 weeks

✓ A. ATT first → ART within 2 weeks

WHO/NACO: Start ATT first (to reduce TB bacillary load), then add ART within 2 weeks for ALL CD4 counts. Earlier ART reduces mortality in severely immunocompromised. Exception: TB MENINGITIS → delay ART 4–8 weeks (paradoxical IRIS risk). Option D is wrong because '8 weeks' is only for meningitis, and that question specifies CD4=250 (no meningitis).

Vector Classification: Water Diseases

Category	Disease Examples
Water-borne (faeco-oral)	Cholera, typhoid, hepatitis A, viral gastroenteritis
Water-washed (poor hygiene)	Scabies, trachoma, bacillary dysentery, louse typhus
Water-based (aquatic invertebrate host)	Schistosomiasis, dracunculiasis (guinea worm)
Water-related (water-breeding vector)	Malaria, filariasis, dengue, YELLOW FEVER, onchocerciasis
Water-dispersed	Legionella pneumophila (cooling towers/water systems)

■ Yellow Fever = Water-RELATED (not water-borne!)

Examiner trap: Yellow fever is vector-borne (Aedes mosquito). Water is needed for Aedes breeding → 'water-related' category. Students confuse this with 'water-borne' (faeco-oral). Yellow fever is NOT transmitted by drinking water.

■ ULTRA HIGH-YIELD REVISION CAPSULE

- ABER = operational efficiency | API = disease burden per 1000 population
- MDR-TB = rifampicin + isoniazid resistance
- GeneXpert = detects MTB + rifampicin resistance in 2 hours
- HIV PEP = TDF+3TC+DTG for 28 days, start <72 hrs (ideally <2 hrs)

- HIV + TB → ATT first → ART within 2 weeks (EXCEPT TB meningitis: delay 4-8 wks)
- Highest vertical HIV risk = maternal viral RNA load
- Secretory IgA (OPV) does NOT cross placenta
- Yellow fever = water-RELATED (water-breeding vector), NOT water-borne
- Berkson's bias = hospital + 2 diseases → inflated association
- NVBDCP: 1 DMC covers 25,000 population

Immunization & Cold Chain

Temperature Zones, VVM, Freeze-Sensitive Vaccines — Exam Favourites

■ What the Examiner Targets

Cold chain questions test SPECIFIC TEMPERATURES for specific vaccines. The #1 tested fact: OPV requires freezing. The #2 tested fact: which vaccines are damaged by freezing (freeze-sensitive). The #3 tested fact: VVM (Vaccine Vial Monitor) interpretation.

Vaccine / Test	Key Temperature / Fact
OPV (Oral Polio Vaccine)	–15 to –25°C (frozen) at state/district level. –20°C most quoted. At PHC/sub-centre: 2–8°C for ≤1 month.
Measles/MMR/Varicella	2–8°C (sensitive to heat AND light). Freeze OK but not needed.
Freeze-sensitive vaccines	DPT, DT, TT, Hep B, Hep A, Rotavirus, IPV — damaged by freezing. Store 2–8°C.
Shake test	Detect freeze-damage in freeze-sensitive vaccines
VVM Stage 1	Inner square LIGHTER than circle → SAFE to use
VVM Stage 2	Square same shade as circle → Use FIRST (still usable but prioritise)
VVM Stage 3	Square DARKER than circle → DO NOT USE
VVM Stage 4	Square much darker → DO NOT USE

Q: Which vaccine is stored at the lowest temperature in the cold chain?

- A. OPV
- B. DPT
- C. Hepatitis B
- D. Rotavirus

✓ **A. OPV (Oral Polio Vaccine)**

OPV must be FROZEN: –15 to –25°C at state/regional vaccine stores (–20°C most quoted). All others (DPT, Hep B, Rota, IPV) = 2–8°C. OPV is the MOST HEAT-SENSITIVE vaccine. Even at 2–8°C it has limited shelf life. Trap: Students think measles vaccine is stored frozen — it's not (2–8°C).

Q: Vaccine vial monitor (VVM) inner square is slightly DARKER than the surrounding circle. Action?

- A. Use the vaccine
- B. Use vaccine but prioritise

C. Discard immediately

D. Shake test needed

✓ **C. DO NOT USE — discard**

VVM darker than circle = Stage 3 or 4 = cumulative heat damage threshold exceeded. DISCARD. Stage 2 (same shade) = 'use FIRST' (still safe but nearing expiry). Stage 1 (lighter) = safely use. Trap: 'use but prioritise' is the Stage 2 instruction, not Stage 3.

Vaccine-Specific Key Facts

Topic	Key Fact
OPV: immunity conferred	Both mucosal (IgA) AND humoral (IgG) — unlike IPV which gives humoral only
Vaccine after disaster	TETANUS (not cholera/typhoid — no mass vaccination evidence post-disaster)
Varicella in seroneg. preg. HCW	VZIG (Varicella-zoster Immunoglobulin) + avoid contact 10 days
Minimum interval: 2 live vaccines	4 weeks (28 days) if given on different days
DTwP: false statement (classic PYQ)	Prior reaction with temp $>37^{\circ}\text{C}$ is NOT a true contraindication ($>40.5^{\circ}\text{C}$ is)
IgA not transplacental	Secretory IgA (OPV) stays in mucosa / breast milk. IgG crosses placenta.
Mission Indradhanush (2014)	Target: children partially/un-immunised. Now covers 12 VPDs.

■ The Cold Chain Ladder — Memorise This Gradient

-20°C → OPV only (frozen zone). $2-8^{\circ}\text{C}$ → ALL other vaccines. Room temperature → maximum 4 hours (open multidose vials, open day policy). Rule: If it must be frozen, it's OPV. If it's harmed by freezing, it's DPT/Hep B/Rota/IPV.

BMW Rules 2016: Colour-Code Recap

Container	Contents
Yellow bag/bin	Human anatomical waste, soiled dressings, blood bags, discarded blood products, cytotoxic/chemical liquid waste
Red bag	Contaminated plastic recyclables (IV sets, syringes without needles, gloves)
White/translucent puncture-proof container	Sharps (needles, blades, glass)
Blue bag/bin	Glass/metallic items, broken ampoules, uncontaminated glass
Black bag	General non-biomedical waste (paper, cardboard — NOT infectious)
Cytotoxic label	Yellow container with 'CYTOTOXIC' label for cytotoxic drugs

■ Blood Bag Goes in YELLOW, Not Red

Trap question: 'Blood bags are disposed in which colour container?' The correct answer is YELLOW (BMW 2016). Red = contaminated PLASTIC recyclables (no blood products). This is asked almost every year in some form.

■ ULTRA HIGH-YIELD REVISION CAPSULE

- OPV = -20°C (lowest temperature in cold chain)
- DPT, Hep B, Rota, IPV = freeze-sensitive ($2-8^{\circ}\text{C}$, damaged by freezing)
- VVM: lighter than circle = use | same shade = use first | darker = DISCARD
- OPV confers mucosal IgA + humoral IgG (IPV only gives humoral IgG)
- Vaccine after disaster = Tetanus (NOT cholera/typhoid)
- 2 live vaccines on different days → minimum 4-week interval
- BMW Yellow = anatomical waste + blood bags + cytotoxic drugs
- BMW Red = contaminated plastic recyclables
- BMW White/translucent = sharps
- Mission Indradhanush = launched 2014, covers 12 VPDs

National Health Programs

JSY, JSSK, RBSK, PMMVY, Mission Indradhanush — Programme Recognition

■ What the Examiner Targets

The examiner tests programme recognition: given a programme's description or features, identify the correct programme. The key is knowing the **BENEFICIARY** and **KEY BENEFIT** of each programme.

Programme	Key Features
JSY (Janani Suraksha Yojana)	Cash incentive for institutional delivery. Rural: ■1400 (high-focus states). Urban: ■1000.
JSSK (Janani Shishu Suraksha Karyakram)	FREE delivery, caesarean, drugs, diagnostics, blood, diet, referral transport for pregnant women + sick neonates.
LaQshya	Labour room & maternity OT quality improvement. Targets respectful maternity care.
Dakshata	Skilled birth attendant (SBA) training. Comprehensive Abortion Care (CAC) training.
SUMAN (Surakshit Matritva Aashwasan)	Zero preventable maternal/newborn deaths. Free delivery, C-section, treatment guarantee.
PMMVY (Pradhan Mantri Matru Vandana Yojana)	■5000 maternity benefit for 1st LIVING child. Conditional cash transfer.
NSSK (Navjaat Shishu Suraksha Karyakram)	Newborn resuscitation training at district/CHC level.
RBSK (Rashtriya Bal Swasthya Karyakram)	Child health screening 0–18 years. 4Ds: Defects, Deficiencies, Diseases, Developmental delays.
NBCC (Newborn Care Corner)	Equipment at every delivery point for immediate newborn care.
Mission Indradhanush	Immunise partially/un-vaccinated children. 12 VPDs.

Reproductive Health Programme (RCH) Structure

Topic	Key Fact
RCH programme implementation level	DISTRICT (differential district approach based on CBR + female literacy)
Sub-centre: coverage	5,000 population (plain) 3,000 (tribal/hilly)
Sub-centre: staff	1 ANM + 1 Male Multipurpose Worker
Sub-centre: most peripheral?	YES — first contact point between PHC system and community
PHC: covers	30,000 population (plain) 20,000 (tribal)
CHC: covers	1,20,000 population (30-bedded)
NUHM: 1 UPHC per	50,000 slum population

Q: The Reproductive and Child Health (RCH) programme activities are targeted at which level?

- A. Sub-centre
- B. Anganwadi
- C. District
- D. Taluka

✓ **C. District**

RCH uses a DIFFERENTIAL approach: all districts categorised A/B/C by crude birth rate and female literacy. Weaker districts get more support. District = planning, resource allocation, and monitoring unit. Trap: Students say 'Sub-centre' because sub-centre delivers maternal/child health services on the ground — but the PROGRAMME is TARGETED at the district level.

ICDS (Integrated Child Development Services)

Aspect	Detail
Beneficiaries	Children 0–6 years + pregnant/lactating women
Implemented through	Anganwadi Centres
Services	Supplementary nutrition, immunisation, health check-up, referral, health education, pre-school education
Nutrition supplement age	6 months to 6 years (complementary feeding)
Complementary feeding start	6 months (WHO recommends 180 days)
WIFS (Weekly Iron Folic Supplementation)	60 mg elemental Fe + 500 µg folic acid WEEKLY for adolescents (10–19 years)

IUD: Contraindications

Category	Examples
ABSOLUTE contraindications	Pregnancy, active/recent PID, undiagnosed vaginal bleeding, cervical/uterine malignancy, previous ectopic pregnancy
RELATIVE contraindications	Uterine malformation (distorted cavity), anaemia, menorrhagia, history of PID
NOT an absolute contraindication	Uterine malformation (only relative)

■ ULTRA HIGH-YIELD REVISION CAPSULE

- JSY = institutional delivery cash incentive | JSSK = free delivery + c-section + transport
- PMMVY = ₹5000 for 1st living child (conditional cash transfer)
- RBSK = 4D screening (0–18 yrs): Defects, Deficiencies, Diseases, Developmental delays
- RCH programme = targeted at DISTRICT level

- Sub-centre = most peripheral health facility (5,000/plain, 3,000/tribal)
- NUHM: 1 UPHC per 50,000 slum population
- WIFS = 60 mg Fe + 500 µg FA weekly for 10–19 year olds
- ICDS beneficiaries = 0–6 years + pregnant/lactating women (not 0–18 years)
- IUD absolute contraindication: NOT uterine malformation (it's only relative)
- Complementary feeding starts at 6 months (180 days)

Environmental & Occupational Health

Water, Air, Waste, Factories Act — Numbers the Examiner Loves

■ What the Examiner Targets

Environmental health is tested through specific NUMBERS and CLASSIFICATIONS. The examiner gives you a scenario and expects you to recall the right law, the right concentration, or the right colour-coded bag.

Water Quality & Purification

Topic	Key Fact
Bleaching powder active agent	Hypochlorous acid (HOCl) — 70-80× more effective than hypochlorite ion
Bleaching powder available chlorine	33% available chlorine
Nalgonda technique	Alum + Lime (defluoridation of drinking water)
Fecal contamination indicator in water	E. coli (most reliable indicator)
Hardness NOT caused by	NaCl (sodium chloride) — hardness caused by Ca^{2+} , Mg^{2+} salts
Water sample during cholera outbreak	Collect BEFORE chlorination (to detect Vibrio, not kill it first)
Fluorosis treatment	Defluoridation (Nalgonda technique) — NOT fluoride supplementation

Q: The active disinfectant property of bleaching powder is due to:

- A. Hypochlorite ion
- B. Chloride ion
- C. Hypochlorous acid
- D. Calcium oxide

✓ **C. Hypochlorous acid (HOCl)**

Bleaching powder (CaOCl_2) releases HOCl on contact with water. HOCl is a weak acid but 70-80x more active as a disinfectant than OCl^- . Trap: Students say 'hypochlorite ion' because chlorine/bleach sounds like hypochlorite. The KEY: it's HYPO-CHLOR-OUS ACID (HOCl), not the hypochlorite ION (OCl^-).

Air Quality Index (AQI)

AQI / Topic	Value / Detail
AQI 0–50	Good (green)
AQI 51–100	Satisfactory (light green)
AQI 101–200	Moderate (yellow)

AQI 201–300	Poor (orange)
AQI 301–400	Very poor (red)
AQI 401–500	Severe (maroon)
AQI = 407 category	Severe (>400)
Routine monitoring indicators	SO ₂ + smoke/PM + Lead (Pb) — Park's classic triad
AQI components (comprehensive)	PM _{2.5} , PM ₁₀ , NO ₂ , SO ₂ , CO, O ₃ , NH ₃ , Pb

Q: AQI of 394–407 is classified as:

- A. Very poor
- B. Hazardous
- C. Poor
- D. Severe

✓ **D. Severe**

AQI 401–500 = Severe (maroon). The range 394–407 spans Very Poor/Severe. 394 = Very Poor; 401+ = Severe. This is a near-boundary question designed to trap. The examiner expects you to know that 401+ = Severe. Note: Some USA scales have 'Hazardous' (>300) but Indian scale uses 'Severe' for 401–500.

Occupational Health: Factories Act 1948

Provision	Detail
Normal working hours	48 hours/week (9 hours/day)
Maximum with overtime	60 hours/week (2 hours overtime/day)
Overtime wages	Double the normal rate
Adolescents (15–18 years)	Maximum 4.5 hours/day, only 6 AM–7 PM
ESI Act: extended sickness TB	Up to 2 years (extended sickness benefit)
ESI Act: normal sickness benefit	91 days/year
PPE: NOT included	Lab coat, radiation badge (monitoring, not protecting)

Q: Maximum work hours per week including overtime under the Factories Act 1948?

- A. 48 hours
- B. 50 hours
- C. 60 hours
- D. 72 hours

✓ C. 60 hours

Normal: 48 h/week. Overtime allowed: up to 2 extra hours/day = maximum 60 h/week. This appears repeatedly in PYQs (2018, 2019, 2020). Trap: '48 hours' is the NORMAL limit; 60 is the MAXIMUM including overtime. The question specifically says 'including overtime' → 60.

Pneumoconiosis: Dust-Disease Matching

Disease	Causative Dust / Occupation
Silicosis	Silica / quartz dust — mining, quarrying, stone cutting
Asbestosis	Asbestos — shipbuilding, insulation, roofing, brake lining
Byssinosis	Cotton dust — textile mills (Monday fever)
Coal worker's pneumoconiosis	Coal dust — coal mining
Siderosis	Iron dust — iron mining/processing (benign)
Anthracosis	Carbon/coal dust — air pollution (common urban finding)
Bagassosis	Sugarcane bagasse — sugar industry
Farmer's lung	Mouldy hay (thermophilic actinomycetes) — hypersensitivity pneumonitis

■ ULTRA HIGH-YIELD REVISION CAPSULE

- Bleaching powder active agent = Hypochlorous acid (HOCl)
- Nalgonda technique = Alum + Lime (defluoridation)
- Factories Act: normal 48 h/wk; max with OT = 60 h/wk
- ESI extended sickness TB = up to 2 years
- AQI 401–500 = Severe | 301–400 = Very Poor | 201–300 = Poor
- BMW Yellow = anatomical waste + chemical liquid waste (cytotoxic + body secretions)
- BMW Red = contaminated plastic recyclables
- BMW White = sharps
- Lab coat = NOT a PPE
- Byssinosis = cotton dust (textile workers)

Health Administration & Infrastructure

Norms, Population Coverage, Administration Levels — Rapid Recall Zone

■ What the Examiner Targets

This chapter is about POPULATION NORMS. The examiner gives a population size and asks what infrastructure serves it, or gives a facility and asks what population it covers.

Facility	Population Coverage / Staff
Sub-centre (plain)	5,000 population 1 ANM + 1 Male MPW
Sub-centre (tribal/hilly)	3,000 population
PHC (plain)	30,000 population 1 Medical Officer
PHC (tribal)	20,000 population
CHC	1,20,000 population 30 beds, 4 specialists
District hospital	1–3 million (district-level)
UPHC (Urban, NUHM)	50,000 slum population
Vision 2020 Primary Vision Centre	50,000 population 1 ophthalmic assistant
Vision 2020 Secondary Service Centre	5 lakh (500,000) population 2 ophthalmologists
Vision 2020 Training Centre	50 lakh (5 million) population
Vision 2020 Centre of Excellence	5 crore (50 million) population

Q: Under Vision 2020 - Right to Sight, a secondary service centre covers which population?

- A. 10,000
- B. 50,000
- C. 1 lakh
- D. 5 lakh

✓ **D. 5 lakh (500,000)**

Vision 2020 pyramid (remember 50k → 5L → 50L → 5Cr): Primary Vision Centre = 50,000 | Secondary Service Centre = 5 LAKH | Training Centre = 50 lakh | Centre of Excellence = 5 crore. Trap: Students confuse Primary Vision Centre (50k) with Secondary Service Centre (5L). The '5' in 5 lakh matches 'Secondary' = both have an 'S' and 5.

Health Administration Concepts

Concept	Key Fact
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Principal administrative unit in India	DISTRICT (headed by District Collector, IAS)
Management by Objective (MBO)	Behavioural science method — goal-setting with employee participation
Cost-benefit analysis	Both costs AND benefits measured in MONETARY terms
Cost-effectiveness analysis	Costs in money; benefits in NATURAL UNITS (DALYs, life-years saved)
Efficiency	Output/Input ratio (how much output for given input)
Effectiveness	Degree to which objectives are achieved (outcome vs goal)
QALY (Quality-Adjusted Life Year)	Combines quantity + quality of life. 1 QALY = 1 year in perfect health.
DALY (Disability-Adjusted Life Year)	YLL (years life lost) + YLD (years lived with disability)
Delphi technique	Structured expert consensus method — repeated anonymous surveys
Clinical management in health programme	Classified under PROCESS (not input or outcome)

ASHA Role

Aspect	Detail
ASHA full form	Accredited Social Health Activist
Coverage	1 per 1000 population (rural village)
Role	Community link worker, mobilisation, facilitates access to health services
Incentives	Performance-based (not fixed salary) — e.g., for institutional deliveries referred
Gray kit	Contains treatment for STIs (syndromic management) — given to ASHA

MHA 2017 — Mental Healthcare Act

Provision	Detail
Voluntary admission max duration	90 days (Section 90)
If admission > 30 days / readmission within 7 days	Two psychiatrists must review + inform MHRB
Consent review	Fortnightly
Advanced Directive (AD)	Patient can specify desired/unwanted treatment in advance

■ ULTRA HIGH-YIELD REVISION CAPSULE

- Sub-centre: 5,000/plain, 3,000/tribal — most peripheral health facility
- PHC: 30,000/plain — 1 Medical Officer
- CHC: 1,20,000 — 30 beds, 4 specialists (FRU = First Referral Unit)

- UPHC (NUHM): 50,000 slum population
- Vision 2020: PVC=50k | Secondary=5 lakh | Training=50L | CoE=5 crore
- DALY = YLL + YLD
- QALY = 1 year in perfect health
- Cost-benefit analysis = both in money | Cost-effectiveness = costs in money + natural units
- MHA 2017 voluntary admission maximum = 90 days
- ASHA = 1 per 1000 population, performance-based incentives

Nutrition, Demography & Vital Statistics

Deficiency Diseases, Vital Rates, Demographic Transition

■ What the Examiner Targets

Nutrition questions test deficiency disease recognition and Vitamin A dosing. Demography questions test which RATE measures what, and what India's demographic stage implies.

Nutritional Deficiency Diseases

Topic	Key Fact
Vitamin A deficiency signs (hierarchy)	Night blindness → Bitot's spots → Xerophthalmia → Keratomalacia
Vitamin A: 2-year-old with keratomalacia	2,00,000 IU on Day 1, Day 2, Day 14 (3 doses)
Vitamin A prophylaxis (6m–5yr)	2,00,000 IU every 6 months
Phrynoderma	Essential Fatty Acid (EFA) deficiency — 'toad skin' (follicular hyperkeratosis)
Neurolathyrism	Lathyrus sativus (khesari dal) → BOAA toxin → spastic paraparesis
Fluorosis (dental/skeletal)	Excess fluoride — treated by NALGONDA defluoridation, NOT supplementation
Iodine RDA (lactating women)	250 µg/day (NIN 2020)
Dietary cholesterol limit (CAD prevention)	< 200 mg/1000 kcal
Gomez Grade I (mild malnutrition)	76–90% weight-for-age
Gomez Grade II (moderate)	61–75% weight-for-age
Gomez Grade III (severe)	< 60% weight-for-age

Q: A child has phrynoderma (rough scaly 'toad skin'). Primarily caused by deficiency of:

- A. Vitamin A
- B. Essential fatty acids
- C. Vitamin D
- D. Vitamin E

✓ **B. Essential fatty acids**

Phrynoderma (follicular hyperkeratosis) is primarily caused by EFA (linoleic acid) deficiency. Students automatically think 'Vitamin A' because Vitamin A causes skin changes (xerosis). Trap: Vitamin A deficiency → xerosis, night blindness, Bitot's spots, keratomalacia — but the specific term PHRYNODERMA points to EFA.

Vital Statistics — Knowing Which Rate Measures What

Rate	Definition / Key Fact
Crude Birth Rate (CBR)	Live births/1000 mid-year population/year. CRUDE = includes non-fertile denominator.
TFR (Total Fertility Rate)	Average children a woman would have in her lifetime. Best measure of fertility trends.
ASFR (Age-Specific Fertility Rate)	Births to women of a specific age group per 1000 women in that age
IMR	Infant deaths (<1 year) per 1000 live births. Best single indicator of community health.
MMR	Maternal deaths per 1,00,000 live births. SDG target: <70 by 2030.
NMR	Neonatal deaths (<28 days) per 1000 live births
CBR disadvantage	Denominator includes males + post-menopausal women → CRUDE, not fertility-specific
Demographic transition (India)	Late expanding/Stage 3 (declining fertility, declining mortality)
CPR = 2.1	Net Reproduction Rate = 1 (replacement fertility level)

Q: CBR is a 'crude' measure because:

- A. It uses a small sample
- B. Denominator includes non-fertile individuals
- C. It is calculated annually
- D. It underestimates actual births

✓ **B. Denominator includes non-fertile individuals**

CBR = live births / total mid-year population × 1000. The denominator includes males, elderly, and children — NOT all are at risk of childbirth. Therefore CBR is CRUDE (approximate). For ACTUAL fertility measurement → TFR or ASFR. TFR is the single best indicator of fertility trends.

Health Indicators: PQLI, HDI, GHI

Indicator	Components / Key Fact
PQLI (Physical Quality of Life Index)	3 components: Life expectancy at age 1 + IMR + Basic literacy rate
HDI (Human Development Index)	3 components: Life expectancy (min 20, max 85 yrs) + Education + GNI per capita
GHI (Global Hunger Index)	4 components: Undernourishment + Child stunting + Child wasting + U5MR (NOT IMR)
GHI: what is NOT included	IMR is NOT a GHI component (U5MR is used instead)
DALY	YLL + YLD (burden of disease measure)
YPLL	Years of Potential Life Lost = age at death – 65. Measures premature deaths.

■ PQLI vs HDI vs GHI — Never Confuse Them

PQLI = 3 components, Morris (1979). Remember 'P for PQLI, P for Physical (child) indicators': Life expectancy at age 1, IMR, Literacy. HDI = 3 components, UNDP. Health + Education + Income. GHI = 4 components, IFPRI. The trap: GHI uses U5MR (under-5 mortality), NOT IMR.

■ ULTRA HIGH-YIELD REVISION CAPSULE

- Vitamin A for keratomalacia: 2,00,000 IU × 3 doses (Day 1, 2, 14)
- Phrynoderma = EFA deficiency (not Vitamin A)
- Gomez: Grade I=76-90% | Grade II=61-75% | Grade III=<60%
- CBR = crude (denominator includes non-fertile) | TFR = best fertility measure
- IMR = best single indicator of community health
- MMR: SDG target <70 per 1,00,000 live births by 2030
- PQLI: life expectancy at 1 yr + IMR + literacy
- GHI: undernourishment + child stunting + child wasting + U5MR (NOT IMR)
- HDI: life expectancy (min 20, max 85 yr) + education + GNI per capita
- CPR = 2.1 → NRR = 1 (replacement level fertility)

Health Indicators & International Health

SDGs, WHO Structure, Vision 2020, Historical Milestones

■ What the Examiner Targets

International health questions test WHO structure, SDG targets, Vision 2020 norms, and historical milestones. These are high-yield **FACTUAL** questions — the answer is usually a very specific number or name.

Topic	Key Fact
SDG 3.1: MMR target by 2030	< 70 per 1,00,000 live births (no country > 140)
SDG 3: Ensure healthy lives	Health-specific SDG. All other health targets under SDG 3.
WHO HQ	Geneva, Switzerland
WHO Regional Office EURO	Copenhagen, Denmark
WHO Regional Office SEARO	New Delhi, India (covers India)
WHO Regional Office PAHO	Washington DC (Americas)
WHO Regional Office EMRO	Cairo (Eastern Mediterranean)
WHO Regional Office AFRO	Brazzaville, Congo
WHO Regional Office WPRO	Manila, Philippines
MONICA project	Multinational MONitoring of trends and determinants In CARDiovascular disease (WHO, 26 countries)
Colombo Plan	Technical/economic cooperation in South and Southeast Asia
First Public Health Act	1848, England — by Edwin Chadwick
Miasma theory	Disease from 'bad air' / rotting matter — pre-germ theory era

Disease Eradication Terminology

Term	Definition / Example
Control	Reduction to acceptable levels. Disease still present.
Elimination	Regional/national interruption of transmission. Causative agent may persist.
Eradication	Worldwide reduction to ZERO cases. Agent may still exist (e.g., variola in labs).
Extinction	Agent completely destroyed worldwide. No longer exists anywhere.
Eradicated diseases	Smallpox (1980). Rinderpest (2011, animal disease).
Eliminated in India	Polio (2014), Guinea worm, Yaws, Neonatal tetanus

Q: Causative agent is present but disease transmission is interrupted in a region. This is:

- A. Eradication
- B. Elimination
- C. Control
- D. Extinction

✓ **B. Elimination**

Elimination = REGIONAL interruption of transmission, even if the organism still exists. Example: India achieved polio ELIMINATION (2014) — poliovirus still exists globally. Eradication = WORLDWIDE zero cases (smallpox, 1980). Extinction = the agent is destroyed everywhere (no natural reservoirs, no lab stocks). Control = reduced but not eliminated.

Indian Constitution: Health-Related Articles

Article / List	Health relevance
Article 21	Right to Life — interpreted to include right to health
Article 47	Duty of State to raise nutrition/health standard
Concurrent List	Both Central AND State can legislate — includes ADULTERATION of food
State List	State-only legislation — includes PUBLIC HEALTH and sanitation
Union List	Central-only — includes ports, quarantine, regulation of medical profession

Disaster Management

Topic	Key Fact
Disaster management cycle sequence	Mitigation → Preparedness → Response → Recovery
Vaccine priority post-disaster	Tetanus (injury + contamination risk). NOT cholera/typhoid.
Toxic fumes (industrial disaster)	Seal windows/doors, stay indoors (shelter-in-place). Do NOT run outside.
PHC disaster preparation	Stockpile drugs, identify manpower, clear communication lines

Surveillance & Monitoring — Final Clarification

Concept	Definition
Surveillance	Continuous systematic collection, analysis, interpretation and dissemination for public health action (includes outbreak detection)
Monitoring	Routine continuous measurement of PROGRAMME INPUTS and OUTPUTS (process measurement)
Evaluation	Measures whether OBJECTIVES have been achieved (outcomes vs goals). Periodic, not continuous.
Sentinel surveillance purpose	Monitor TRENDS in high-risk or hard-to-reach populations at selected sites

De facto census	Counted at place of ENUMERATION (wherever found on census date)
De jure census	Counted at place of USUAL RESIDENCE (not where found)

■ ULTRA HIGH-YIELD REVISION CAPSULE

- SDG 3.1: MMR < 70/1,00,000 live births by 2030
- WHO SEARO = New Delhi | EURO = Copenhagen | HQ = Geneva
- MONICA = cardiovascular disease trends monitoring (WHO, 26 countries)
- Eradication = worldwide zero | Elimination = regional interruption | Extinction = agent destroyed
- Smallpox = only human disease eradicated (1980)
- India polio elimination = 2014
- Concurrent List includes food adulteration | State List includes public health
- Disaster cycle: Mitigation → Preparedness → Response → Recovery
- Monitoring = programme process (inputs/outputs) | Evaluation = outcomes vs goals
- Sentinel surveillance = high-risk/hard-to-reach populations at selected sites

You Have Reached the End — Now the Real Work Begins

PSM is not about memorising 260 facts. It is about recognising 15 patterns that the examiner uses to construct those questions. You have now seen all 15. The next step is ACTIVE RECALL — close the book, take a blank paper, and try to reconstruct each chapter's revision capsule from memory. Every item you miss is a revision priority.

Good luck. The exam is yours to own.

Based on 260+ NEET-PG PYQs (2015–2025) from the Neet-Pg2026 Question Bank Repository.
For errors or updates, contribute to: github.com/skm9097/neet-pg2026